

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250268143

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

Pereira de Camargo; Jean Carlos et al.

SYSTEM AND METHOD FOR AN AGRICULTURAL HARVESTER

Abstract

An elevator assembly for a harvester may include an elevator housing and an elevator extending within the elevator housing between a proximal end portion and a distal end portion. The elevator may be configured to carry harvested crops between the proximal end portion and the distal end portion. A discharge assembly may be operably coupled with the elevator housing. The discharge assembly may include a discharge housing defining a discharge outlet, a deflector operably coupled with the discharge housing, and a discharge actuator operably coupled with the elevator housing and the discharge housing.

Inventors: Pereira de Camargo; Jean Carlos (Piracicaba-SP, BR), Moreno; Anderson Antonio (Piracicaba-SP, BR), Pereira; Alessandro Roberto (Piracicaba-SP, BR), Pereira; Daniel Rohrer (Piracicaba-SP, BR)

Applicant: CNH Industrial Brasil Ltda. (Nova Lima, BR)

Family ID: 1000008487130

Appl. No.: 19/062766

Filed: February 25, 2025

Foreign Application Priority Data

BR 10 2024 003769 3

Feb. 26, 2024

Publication Classification

Int. Cl.: A01D61/02 (20060101); B65G47/44 (20060101)

U.S. Cl.:

CPC A01D61/02 (20130101); B65G47/44 (20130101);

Background/Summary

FIELD OF THE INVENTION

[0001] The present disclosure relates generally to agricultural harvesters, such as sugarcane harvesters, and, more particularly, to systems and methods for an elevator assembly for an agricultural harvester.

BACKGROUND OF THE INVENTION

[0002] Agricultural harvesters can include an assembly of processing components for processing harvested material. For instance, within a sugarcane harvester, severed sugarcane stalks are conveyed via a feed roller assembly to a chopper assembly that cuts or chops the sugarcane stalks into pieces or billets (e.g., six-inch cane sections). The processed harvested material discharged from the chopper assembly is then directed as a stream of billets and debris into a primary extractor, within which the airborne debris (e.g., dust, dirt, leaves, etc.) is separated from the sugarcane billets. The separated/cleaned billets then fall into an elevator assembly for delivery to an external storage device. However, in some instances, the billets may be directed to a common section of the storage device.

[0003] Additionally, when an external receiver is absent or is otherwise not properly positioned relative to the harvester, the elevator assembly may be stopped to prevent the conveyed billets from being discharged onto the ground. This situation can arise under a variety of conditions, such as if the accompanying receiver is full. As another example, the receiver may often be a towed wagon that (along with its towing vehicle) defines a larger turning radius than the harvester itself. In such instances, when a turn is being executed at the end of the field, the receiver may not be immediately present to receive the harvested crops. As a result, the harvester may have to pause operation until the receiver can be properly positioned relative to the harvester. In either situation, there is a loss in the productivity of the harvester.

[0004] Accordingly, an improved elevator assembly for an agricultural harvester would be welcomed in the technology.

BRIEF DESCRIPTION OF THE INVENTION

[0005] Aspects and advantages of the invention will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the invention.

[0006] In some aspects, the present subject matter is directed to an elevator assembly for a harvester that includes an elevator housing and an elevator extending within the elevator housing between a proximal end portion and a distal end portion. The elevator is configured to carry harvested crops between the proximal end portion and the distal end portion. A discharge assembly is operably coupled with the elevator housing. The discharge assembly includes a discharge housing defining a discharge outlet. The discharge housing is movable relative to the elevator housing. A deflector is operably coupled with the discharge housing.

[0007] In some aspects, the present subject matter is directed to a method for operating a harvester. The harvester can include an elevator assembly including an elevator extending between a proximal end portion and a distal end portion and a discharge assembly. The method includes conveying a material along the elevator in a direction from the proximate end portion to the distal end portion. The method also includes altering, with a discharge actuator, a position of a discharge housing of the discharge assembly relative to an elevator housing, the discharge assembly downstream of the elevator. Lastly, the method includes rotating a deflector operably coupled with the discharge housing.

[0008] In some aspects, the present subject matter is directed to an elevator assembly for a harvester that includes an elevator housing and an elevator configured to carry material there along.

A discharge assembly is operably coupled with the elevator housing. The discharge assembly includes a discharge housing defining a discharge outlet. The discharge housing is movable relative to the elevator housing. A discharge actuator is operably coupled with the elevator housing and the discharge housing. The discharge actuator is configured to alter a position of the discharge housing relative to the elevator housing.

[0009] These and other features, aspects, and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:

[0011] FIG. 1A illustrates a simplified, side view of an agricultural harvester in accordance with aspects of the present subject matter;

[0012] FIG. 1B illustrates a simplified, side view of an agricultural harvester in accordance with aspects of the present subject matter;

[0013] FIG. 2 is a rear perspective view of an elevator assembly of the agricultural harvester in accordance with aspects of the present subject matter;

[0014] FIG. 3 is a cross-sectional view of the elevator assembly of FIG. 2 taken along the line III-III;

[0015] FIG. 4 is a side plan view of the elevator assembly with an ejection assembly housing and flap in a first position in accordance with aspects of the present subject matter;

[0016] FIG. 5 is a side plan view of the elevator assembly with an ejection assembly housing and flap in a second position in accordance with aspects of the present subject matter;

[0017] FIG. 6 is a side plan view of the elevator assembly with an ejection assembly housing and flap in a third position in accordance with aspects of the present subject matter; and

[0018] FIG. 7 illustrates a flow diagram of a method for agricultural harvesting in accordance with aspects of the present subject matter.

[0019] Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present technology.

DETAILED DESCRIPTION OF THE INVENTION

[0020] Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0021] In this document, relational terms, such as first and second, top and bottom, and the like, are used solely to distinguish one entity or action from another entity or action, without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of

elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.

[0022] As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify a location or importance of the individual components. The terms “coupled,” “fixed,” “attached to,” and the like refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or more intermediate components or features, unless otherwise specified herein. The terms “upstream” and “downstream” refer to the relative direction with respect to a harvested material along a path. For example, “upstream” refers to the direction from which a harvested material flows, and “downstream” refers to the direction to which the harvested material moves. The term “selectively” refers to a component's ability to operate in various states (e.g., an ON state and an OFF state) based on manual and/or automatic control of the component.

[0023] Furthermore, any arrangement of components to achieve the same functionality is effectively “associated” such that the functionality is achieved. Hence, any two components herein combined to achieve a particular functionality can be seen as “associated with” each other such that the desired functionality is achieved, irrespective of architectures or intermedial components. Likewise, any two components so associated can also be viewed as being “operably connected” or “operably coupled” to each other to achieve the desired functionality, and any two components capable of being so associated can also be viewed as being “operably couplable” to each other to achieve the desired functionality. Some examples of operably couplable include, but are not limited to, physically mateable, physically interacting components, wirelessly interactable, wirelessly interacting components, logically interacting, and/or logically interactable components.

[0024] The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

[0025] Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” “generally,” and “substantially,” is not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or apparatus for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a ten percent margin.

[0026] Moreover, the technology of the present application will be described in relation to exemplary embodiments. The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments. Additionally, unless specifically identified otherwise, all embodiments described herein will be considered exemplary.

[0027] As used herein, the term “and/or,” when used in a list of two or more items, means that any one of the listed items can be employed by itself, or any combination of two or more of the listed items can be employed. For example, if a composition or assembly is described as containing components A, B, and/or C, the composition or assembly can contain A alone; B alone; C alone; A and B in combination; A and C in combination; B and C in combination; or A, B, and C in combination.

[0028] As used herein, “harvested material” can include crop product, which may be in the form of stalks (or to be collected crop or after collected crop), billets (modified stalks or modified collected crop), and/or debris, which may be any object other than the stalks or the billets (or the collected crop) (e.g., dust, dirt, leaves, etc.).

[0029] In general, the present subject matter is directed to a system for an agricultural harvester.

The system can include an elevator assembly for a harvester that may include an elevator housing and an elevator extending within the elevator housing between a proximal end portion and a distal end portion. The elevator may be configured to carry harvested crops between the proximal end portion and the distal end portion.

[0030] The elevator assembly may include and/or be operably coupled with a discharge assembly that is operably coupled with the elevator housing. The discharge assembly may include a discharge housing defining a discharge outlet. In some cases, the discharge housing may be movable relative to the elevator housing. A deflector may be operably coupled with the discharge housing. A discharge actuator may be operably coupled with the elevator housing and the discharge housing. The discharge actuator may be configured to alter a position of the discharge housing relative to the elevator housing. In some instances, the discharge assembly may be configured to receive inputs associated with a desired operating mode for the harvester **10** and/or a discharge location and, in turn, adjust a position of the discharge housing relative to the elevator housing.

[0031] Referring now to the drawings, FIGS. **1A** and **1B** illustrate a harvester **10** configured as a sugarcane harvester. However, in other examples, the harvester **10** may correspond to any other suitable agricultural harvester. As shown in FIGS. **1A** and **1B**, the harvester **10** can include a frame **12**, a pair of front wheels **14**, a pair of rear wheels **16**, and an operator's cab **18**. The harvester **10** may also include a power source **20** (e.g., an engine mounted on the frame **12**) that powers one or both pairs of the wheels **14**, **16** via a driveline assembly **22** (e.g., a transmission) to traverse a field **24**. Alternatively, the harvester **10** may be a track-driven harvester and, thus, may include tracks driven by the power source **20** as opposed to the illustrated wheels **14**, **16**. The power source **20** may also drive a hydraulic fluid pump **26** configured to generate pressurized hydraulic fluid for a hydraulic circuit, which may be configured to power various components of the harvester **10**, including the driveline assembly **22**.

[0032] The harvester **10** may also include a material processing system **28** incorporating various components, assemblies, and/or sub-assemblies of the harvester **10** for cutting, processing, cleaning, and discharging sugarcane as the cane is harvested from an agricultural field **24**. For instance, the material processing system **28** may include a topper assembly **30** positioned at the front end portion of the harvester **10** to intercept sugarcane as the harvester **10** is moved in a forward direction. As shown, the topper assembly **30** may include both a gathering disk **32** and a cutting disk **34**. The gathering disk **32** may be configured to gather the sugarcane stalks **60S** so that the cutting disk **34** may be used to cut off the top of each stalk **60S**. As is generally understood, the height of the topper assembly **30** may be adjustable via a pair of arms **36**, which may be hydraulically raised and lowered.

[0033] The material processing system **28** may further include a crop divider **38** that extends upwardly and rearwardly from the field **24**. In general, the crop divider **38** may include two spiral feed rollers **40**. Each feed roller **40** may include a ground shoe **42** at its lower end portion to assist the crop divider **38** in gathering the sugarcane stalks **60S** for harvesting. Moreover, as shown in FIGS. **1A** and **1B**, the material processing system **28** may include a knock-down roller **44** positioned near the front wheels **14** and a fin roller **46** positioned behind the knock-down roller **44**. As the knock-down roller **44** is rotated, the sugarcane stalks **60S** being harvested are knocked down while the crop divider **38** gathers the stalks **60S** from agricultural field **24**. Further, as shown in FIGS. **1A** and **1B**, the fin roller **46** may include a plurality of intermittently mounted fins **48** that assist in forcing the sugarcane stalks **60S** downward.

[0034] Referring still to FIGS. **1A** and **1B**, the material processing system **28** of the harvester **10** may also include a base cutter assembly **50** positioned behind the fin roller **46**. The base cutter assembly **50** may include blades for severing the sugarcane stalks **60S** as the cane is being harvested. The blades, which may be located on a periphery section of the base cutter assembly **50**, can be rotated by the hydraulic circuit. Additionally, in several examples, the blades may be angled downward to sever the base of the sugarcane as the cane is knocked down by the fin roller **46**.

[0035] Moreover, the material processing system **28** may include a feed roller assembly **52** located downstream of the base cutter assembly **50** for moving the severed stalks **60S** of sugarcane from base cutter assembly **50** along the processing path of the material processing system **28**. As shown in FIGS. **1A** and **1B**, the feed roller assembly **52** may include a plurality of bottom rollers **54** and a plurality of opposed, top rollers **56**. The various bottom and top rollers **54**, **56** may be used to pinch the harvested sugarcane during transport. As the sugarcane is transported through the feed roller assembly **52**, debris **64** (e.g., rocks, dirt, and/or the like) may be allowed to fall through bottom rollers **54** onto the field **24**.

[0036] In addition, the material processing system **28** may include a chopper assembly **58** located at the downstream end section of the feed roller assembly **52** (e.g., adjacent to the rearward-most bottom roller **54** and the rearward-most top roller **56**). In general, the chopper assembly **58** may be used to cut or chop the severed sugarcane stalks **60S** into pieces or “billets” **60B**, which may be, for example, six (6) inches long. The billets **60B** may then be propelled towards an elevator assembly **62** of the material processing system **28** for delivery to an external receiver or storage device.

[0037] The debris **64** (e.g., dust, dirt, leaves, etc.) separated from the sugarcane billets **60B** may be expelled from the harvester **10** through a primary extractor **66** of the material processing system **28**, which may be located downstream of the chopper assembly **58** and may be oriented to direct the debris **64** outwardly from the harvester **10**. Additionally, an extractor fan **68** may be mounted within an extractor housing **70** of the primary extractor **66** for generating a suction force or vacuum sufficient to force the debris **64** through the primary extractor **66**. The separated or cleaned billets **60B**, which may be heavier than the debris **64** expelled through the extractor **66**, may then fall downward to the elevator assembly **62**.

[0038] As shown in FIGS. **1A** and **1B**, the elevator assembly **62** may include an elevator housing **72** and an elevator **74** extending within the elevator housing **72** between a lower, proximal end portion **76** and an upper, distal end portion **78**. In some examples, the elevator **74** may include a looped chain **80** and a plurality of flights or paddles **82** attached to and spaced on the chain **80**. The paddles **82** may be configured to hold the sugarcane billets **60B** on the elevator **74** as the sugarcane billets **60B** are elevated along a top surface of the elevator **74** defined between its proximal and distal end portions **76**, **78**. A region **84** for retaining the harvested material may be defined between the first and second paddles **82** operably coupled with the elevator **74**. As such, a first region **84A** may be defined between first and second paddles **82**, a second region **84B** may be defined between the second and a third paddle **82**, and so on. Additionally, the elevator **74** may include lower and upper sprockets **86A**, **86B** positioned at its proximal and distal end portions **76**, **78**, respectively. In some instances, the elevator **74** may include a first span **88**, a second span **90** downstream of the first span **88**, and a third span **92** downstream of the first span **88** and the second span **90**. In such examples, the second span **90** can be offset from the first span **88** and the third span **92** may be offset from the first span **88** and/or the second span **90**. In some cases, an intermediate sprocket **86C** can be positioned between the first span **88** and the second span **90** and between the second span **90** and the third span **92**.

[0039] As shown in FIGS. **1A** and **1B**, an elevator motor **94** may be coupled to one of the sprockets (e.g., the upper sprocket **86B**) for driving the chain **80**, thereby allowing the chain **80** and the paddles **82** to travel in a loop between the proximal end portions **76** and the distal end portion **78** of the elevator **74**.

[0040] Moreover, in some examples, debris **64** (e.g., dust, dirt, leaves, etc.) separated from the elevated sugarcane billets **60B** may be expelled from the harvester **10** through a secondary extractor **96** of the material processing system **28** coupled to the rear end portion of the elevator housing **72**. For example, the debris **64** expelled by the secondary extractor **96** may be debris **64** remaining after the billets **60B** are cleaned and debris **64** expelled by the primary extractor **66**. As shown in FIGS. **1A** and **1B**, the secondary extractor **96** may be located adjacent to the distal end portion **78** of the elevator **74** and may be oriented to direct the debris **64** outwardly from the

harvester **10**. Additionally, an extractor fan **98** may be mounted within an extractor housing **100** of the secondary extractor **96** to generate a suction force or vacuum sufficient to force the debris **64** through the secondary extractor **96**. The separated, cleaned billets **60B**, heavier than the debris **64** expelled through the primary extractor **66**, may then fall from the distal end portion **78** of the elevator **74**.

[0041] In some instances, the billets **60B** may fall through a discharge outlet **102** defined by a discharge assembly **104** into an external storage device, such as a sugarcane billet cart. In some instances, the discharge assembly **104** can include a discharge housing **106** that may be movable, pivotable, and/or translatable relative to the elevator housing **72**. In various instances, the elevator housing **72** may define an elevator opening **108** and the discharge housing **106** may be positioned at least partially within the elevator opening **108**. In some cases, a deflector **110** may be operably coupled with the discharge housing **106** and configured to prevent the motion of the billets **60B** beyond the deflector **110**.

[0042] In some cases, the discharge assembly **104** can further include a discharge actuator **112** configured to move the discharge housing **106** between various positions. In general, the discharge actuator **112** may correspond to any suitable actuation device. For instance, the discharge actuator **112** may correspond to a linear actuator, such as a fluid-driven cylinder actuator or an electric actuator (e.g., a solenoid-activated actuator). In such instances, the discharge actuator **112** may include an actuator housing **114** and a drive rod **116**. In operation, by linearly actuating the drive rod **116** in one direction or the other, the discharge housing **106** may be moved relative to the elevator housing **72**. In various examples, the actuator housing **114** may be operably coupled to the elevator assembly **62** and/or any other component of the harvester **10**, and the drive rod **116** may be operably coupled to a portion of the discharge housing **106**. Alternatively, the actuator housing **114** may be operably coupled to a portion of the discharge housing **106** and the drive rod **116** may be operably coupled to the elevator assembly **62** and/or any other component of the harvester **10**. Additionally or alternatively, the discharge actuator **112** may correspond to any other suitable actuation device, such as any other suitable linear actuator (e.g., a gear and rack assembly).

[0043] Referring further to FIG. **1B**, in some examples, the elevator assembly **62** may also include a storage hopper **118** at a location adjacent to the distal end portion **78** of the elevator **74** (e.g., at a location below the distal end portion **78** of the elevator **74** and the secondary extractor **96**). As shown in FIG. **1B**, the storage hopper **118** may include a conveyor **120** configured to carry the billets **60B** received from the elevator **74** toward the discharge outlet **102** for ejection from the harvester **10**. In this regard, when the harvester **10** is operated in an unloading mode, the conveyor **120** may be driven in a manner that carries the billets **60B** expelled from the distal end portion **78** of the elevator **74** toward the discharge outlet **102** for ejection from the harvester **10**. The ejected billets **60B** may fall into an associated external receiver or storage device. However, when the harvester **10** is being operated in a storage harvesting mode, the speed of the conveyor **120** may be reduced or the operation of the conveyor **120** may be halted. As such, the billets **60B** expelled from the distal end portion **78** of the elevator **74** may accumulate within a storage volume **122** defined by the storage hopper **118** for temporary storage therein.

[0044] In several examples, the conveyor **120** may extend within the storage hopper **118** between a first, forward end portion and a second, rear end portion. For example, as shown, the first end portion of the conveyor **120** may be positioned adjacent to the distal end portion **78** of the elevator **74** and a forward wall of the storage hopper **118**, while the second end portion of the conveyor **120** may be positioned adjacent to a rear side of the hopper **118**. Moreover, as shown, the conveyor **120** may be positioned vertically below the distal end portion **78** of the elevator **74**. Additionally, the conveyor **120** may include a conveyor belt **124** and forward and rear rollers **126**, **128** positioned at its first and second end portions, respectively. In this regard, a conveyor motor **130** (e.g., a hydraulic motor, electric motor, and/or any other practicable device) may be coupled to one of the rollers (e.g., the rear roller **126**) for driving the conveyor belt **124**, thereby allowing the belt **124** to

travel in an endless loop between the first and second end portions of the conveyor **120**. In such instances, the billets **60B** discharged from the elevator **74** may fall onto a top span of the conveyor belt **124**. Such billets **60B** may, in turn, be carried or otherwise transported from the first end portion of the conveyor **120** to the second end portion of the conveyor **120** as the conveyor belt **124** is moved relative to the rollers **126**, **128**.

[0045] During operation, the harvester **10** traverses the agricultural field **24** for harvesting sugarcane. After the height of the topper assembly **30** is adjusted via the arms **36**, the gathering disk **32** on the topper assembly **30** may function to gather the sugarcane stalks **60S** as the harvester **10** proceeds across the field **24**, while the cutting disk **34** severs the leafy tops of the sugarcane stalks **60S** for disposal along either side of harvester **10**. As the stalks **60S** enter the crop divider **38**, the ground shoes **42** may set the operating width to determine the quantity of sugarcane entering the throat of the harvester **10**. The spiral feed rollers **40** then gather the stalks **60S** into the throat to allow the knock-down roller **44** to bend the stalks **60S** downwardly in conjunction with the action of the fin roller **46**. Once the stalks **60S** are angled downward as shown in FIGS. **1A** and **1B**, the base cutter assembly **50** may then sever the base of the stalks **60S** from field **24**. The severed stalks **60S** are then, by the movement of the harvester **10**, directed to the feed roller assembly **52**.

[0046] The severed sugarcane stalks **60S** are conveyed rearwardly by the bottom and top rollers **54**, **56**, which compresses the stalks **60S**, makes them more uniform, and shakes loose debris **64** to pass through the bottom rollers **54** to the field **24**. At the downstream end portion of the feed roller assembly **52**, the chopper assembly **58** cuts or chops the compressed sugarcane stalks **60S** or harvested material into pieces or billets **60B** (e.g., 6-inch cane sections). The processed harvested material discharged from the chopper assembly **58** is then directed as a stream of billets **60B** and debris **64** into the primary extractor **66**. The airborne debris **64** (e.g., dust, dirt, leaves, etc.) separated from the billets **60B** is then extracted through the primary extractor **66** using suction created by the extractor fan **68**. The separated/cleaned billets **60B** then be directed to an elevator hopper **118** into the elevator assembly **62** and travel upwardly via the elevator **74** from its proximal end portion **76** to its distal end portion **78**.

[0047] During an unloading mode, once the billets **60B** reach the distal end portion **78** of the elevator **74**, the billets **60B** fall through the elevator opening **108** to an external storage device. If provided, the secondary extractor **96** (with the aid of the extractor fan **98**) blows out trash/debris **64** from the harvester **10**, similar to the primary extractor **66**. As provided herein, the discharge assembly **104** may be placed in various positions to direct the billets **60B** in a defined direction such that the ejected billets **60B** may fall into an associated external receiver or storage device. Additionally or alternatively, the harvester **10** may be operated in a storage harvesting mode. In such instances, the speed of the conveyor **120** may be reduced or the operation of the conveyor **120** may be halted, if a conveyor **120** is installed within the elevator assembly **62**. In addition, the discharge housing **106** and/or the deflector **110** may be rotated such that the billets **60B** are retained within the hopper **118**. In some cases, the deflector **110** may be rotated relative to the discharge housing **106** through a deflector actuator **132** (e.g., a hydraulic motor, electric motor, and/or any other practicable device) that may be coupled to the discharge housing **106** (and/or any other component of the harvester **10**). In some instances, the deflector actuator **132** may be configured to alter a position of the deflector **110** relative to the discharge housing **106** between a first position in which at least a portion of the discharge outlet **102** is blocked by the deflector **110** to define a storage harvesting mode and a second position in which at least a portion of the deflector **110** is positioned vertically below the discharge housing **106**.

[0048] Referring back to FIGS. **1A** and **1B**, in several examples, the operation of the discharge actuator **112**, the deflector actuator **132**, and/or the conveyor motor **130** may be controlled through one or more input devices **134**. In various examples, the one or more input devices **134** may include a sensor system **136**, which may include one or more sensors configured to monitor various conditions (e.g., a position of an external storage device, a fill level of the hopper **118**, and/or any

other condition), one or more user interfaces **138** for allowing operator inputs to be provided (e.g., buttons, knobs, dials, levers, joysticks, touch screens, and/or the like), one or more other internal data sources associated with the harvester **10** (e.g., other devices, databases, etc.), one or more external data sources (e.g., a remote computing device or server), and/or any other suitable input devices **134**.

[0049] As shown in FIGS. **1A** and **1B**, a computing system **140** may be operably coupled with the one or more input devices **134**, which may be positioned within the cab **18** and/or in any other practicable location, via one or more communicative links **142**, such as a wired connection and/or a wireless connection. Additionally or alternatively, the computing system **140** may also be operably coupled with the discharge actuator **112**, the deflector actuator **132**, and/or the conveyor motor **130** via one or more communicative links **142**, such as a wired connection and/or a wireless connection. In examples in which the discharge actuator **112**, the deflector actuator **132**, and/or the conveyor motor **130** corresponds to a fluid-driven component(s), the computing system **140** may, instead, be communicatively coupled to suitable electronically controlled valves and/or other suitable fluid-related components for controlling the operation of the discharge actuator **112**, the deflector actuator **132**, and/or the conveyor motor **130**. Regardless, by providing the disclosed communicative links **142** between the computing system **140** and the discharge actuator **112**, the deflector actuator **132**, and/or the conveyor motor **130**, the computing system **140** may be configured to control the operation of the discharge actuator **112**, the deflector actuator **132**, and/or the conveyor motor **130**. For instance, the computing system **140** may be configured to receive inputs associated with the desired operating mode for the harvester **10** and/or a discharge location. In such instances, the computing system **140** may receive an input indicating the desire to switch the operation of the harvester **10** from the discharge harvesting mode to the storage harvesting mode. In turn, the computing system **140** may be configured to electronically control the operation of the conveyor motor **130** to reduce the speed of and/or halt the operation of the conveyor **120** and electronically control the discharge assembly **104** to move to its closed position (via rotation of the discharge housing **106** and/or the deflector **110**). Similarly, if the computing system **140** receives an input indicating the desire to switch the operation of the harvester **10** from the storage harvesting mode back to the discharge harvesting mode, the computing system **140** may be configured to electronically control the operation of the conveyor motor **130** such that the conveyor belt **124** is driven at the desired or selected speed and electronically control the discharge assembly **104** to move to its opened position (via rotation of the discharge housing **106** and/or the deflector **110**). Additionally or alternatively, if the computing system **140** receives an input indicating a change in discharge location (e.g., to fill a different section of a storage container, based on a change in position of the storage container, and/or for any other purpose), the computing system **140** may be configured to electronically control the operation of the discharge actuator **112** to change a position of the discharge housing **106** relative to the elevator housing **72** and/or rotate the deflector **110** relative to the discharge housing **106**.

[0050] In general, the computing system **140** may correspond to any suitable processor-based device known in the art, such as a computing device or any suitable combination of computing devices. Thus, in several examples, the computing system **140** may include one or more processor(s) **144** and associated memory device(s) **146** configured to perform a variety of computer-implemented functions. As used herein, the term “processor” refers not only to integrated circuits referred to in the art as being included in a computer, but also refers to a computing system, a microcontroller, a microcomputer, a programmable logic controller (PLC), an application specific integrated circuit, and other programmable circuits. Additionally, the memory device(s) **146** of the computing system **140** may generally comprise memory element(s) including, but not limited to, computer readable medium (e.g., random access memory (RAM)), computer readable non-volatile medium (e.g., a flash memory), a compact disc-read only memory (CD-ROM), a magneto-optical disk (MOD), a digital versatile disc (DVD) and/or other suitable memory elements. Such memory

device(s) **146** may generally be configured to store suitable computer-readable instructions that, when implemented by the processor(s) **144**, configure the computing system **140** to perform various computer-implemented functions, such as one or more aspects of the method **200** described below with reference to FIG. 7. In addition, the computing system **140** may also include various other suitable components, such as a communications circuit or module, one or more input/output channels, a data/control bus and/or the like.

[0051] In various examples, the computing system **140** may correspond to an existing computing system of the harvester **10**, or the computing system **140** may correspond to a separate processing device. For instance, in one example, the computing system **140** may form all or part of a separate plug-in module that may be installed within the harvester **10** to allow the present subject matter to be implemented without requiring additional software to be uploaded onto existing control devices of the harvester **10**.

[0052] Additionally, the computing system **140** may be configured to electronically control any other suitable components of the harvester **10** in addition to the above-described components. For instance, as shown in FIGS. **1A** and **1B**, the computing system **140** may be communicatively coupled to the elevator motor **94** for controlling the operation of the elevator **74**. For instance, the computing system **140** may electronically control the operation of the elevator motor **94** to automatically adjust the operational speed of the elevator **74** and/or the start/stop the elevator **74**, as desired.

[0053] Referring now to FIGS. **2** and **3**, the discharge housing **106** may include a top portion **150**, a first side portion **152**, a second side portion **154**, and a bottom portion **156**, which, in combination, define the discharge outlet **102**. The housing may further define a bounded area, which may be the hopper **118** downstream of the elevator **74**. The discharge housing **106** may further define a mounting portion **158**. A deflector bracket **160** may be operably coupled with the mounting portion **158**. The deflector bracket **160** may be operably coupled with the deflector **110** to allow the deflector **110** to rotate or otherwise move relative to the discharge housing **106**.

[0054] As illustrated, the deflector **110** can include a central section **162**, a first lateral end section **164**, and a second lateral end section **166**. In some cases, when the deflector **110** is rotated to a first position relative to the discharge housing **106**, the first lateral end section **164** may be positioned on an opposing side of the first side portion **152** of the discharge housing **106** from the bounded area and the second lateral end section **166** may be positioned on an opposing side of the second side portion **154** of the discharge housing **106** from the bounded area. Moreover, the central section **162** may cover at least a portion of the discharge outlet **102**.

[0055] As provided herein, the discharge housing **106** may be movable relative to the elevator housing **72**. In some cases, the discharge housing **106** may include a member **168** and the elevator housing **72** may include a guide **170** (or vice versa). In such instances, the guide **170** may direct the movement of the member **168** along the guide **170**, which, in turn, can direct the movement of the discharge housing **106** relative to the elevator housing **72**.

[0056] In some examples, movement of the discharge housing **106** relative to the elevator housing **72** may be accomplished with the assistance of the discharge actuator **112**. As illustrated, the housing of the discharge actuator **112** may be operably coupled with a first link **172**. The first link **172** may further be operably coupled with the elevator housing **72**. Additionally or alternatively, the drive rod **116** of the discharge actuator **112** may be operably coupled with a second link **174**. The second link **174** may further be operably coupled with the discharge housing **106**. Any of the connections (e.g., the first link **172** to the elevator housing **72**, the first link **172** to the discharge actuator **112**, the second link **174** to the discharge actuator **112**, and the second link **174** to the discharge housing **106**) may be fixed or pivotable in various implementations. In various examples, the discharge actuator **112** may be directly coupled with the elevator housing **72** and/or the discharge housing **106** without the use of a link. In addition, as provided herein, the housing of the discharge actuator **112** may be operably coupled with the elevator housing **72** and the drive rod **116**

may be operably coupled with the discharge housing **106**, or vice versa.

[0057] Referring now to FIGS. **4-6**, in various examples, the elevator **74** within the harvester **10** may be of varied lengths. As such, a distance between the end portion of the elevator **74** and the elevator assembly **62** may also be varied. In various cases, the position of the discharge housing **106** relative to the elevator housing **72** may be at least partially based on the length of the elevator **74** within the elevator assembly **62**. For instance, as shown in FIG. **4**, a first elevator **74** may be positioned within the elevator assembly **62** and a first distance $d_{sub.1}$ may be defined between the end portion of the elevator **74** and the rear wall of the elevator housing **72**. Additionally or alternatively, as shown in FIG. **5**, a second elevator **74** may be positioned within the elevator assembly **62** and a second distance d_e may be defined between the end portion of the elevator **74** and the rear wall of the elevator housing **72**. Additionally or alternatively, as shown in FIG. **6**, a third elevator **74** may be positioned within the elevator assembly **62** and a third distance $d_{sub.3}$ may be defined between the end portion of the elevator **74** and the rear wall of the elevator housing **72**. Based on the distance (e.g., the first distance $d_{sub.1}$, the second distance $d_{sub.2}$, and/or the third distance $d_{sub.3}$), the discharge housing **106** may be positioned in a defined position relative to the elevator housing **72**.

[0058] Additionally, as provided herein, to generally fill the storage container, and/or for any other reason, a discharge location may be varied. In such instances, the position of the discharge housing **106** relative to the elevator housing **72** may be altered. The discharge location may be based on data received from a sensor system **136** (onboard or remote from the harvester **10**), received through a user input, and/or through any other manner. For instance, when the billets **60B** are to be ejected in a first direction, a first offset angle $\theta_{sub.1}$ may be defined between the top portion **150** of the discharge housing **106** and a rearward portion of the elevator housing **72**. When the billets **60B** are to be ejected in a second direction, a second offset angle $\theta_{sub.2}$ may be defined between the top portion **150** of the discharge housing **106** and a rearward portion of the elevator housing **72**. Likewise, when the billets **60B** are to be ejected in a third direction, a third offset angle $\theta_{sub.3}$ may be defined between the top portion **150** of the discharge housing **106** and a rearward portion of the elevator housing **72**. Each of the first angle $\theta_{sub.1}$, the second angle $\theta_{sub.2}$, and the third angle $\theta_{sub.3}$ may be varied from one another. Moreover, in some cases, the deflector **110** may be pivotably mounted to the discharge housing **106** and adjusted based on the position of the discharge housing **106**. Additionally or alternatively, as provided herein, the position of the deflector **110** may be controlled independently of the discharge housing **106**. In such instances, the deflector **110** may be positioned in any position as well based on the received inputs and/or for any other purpose.

[0059] Referring now to FIG. **7**, a flow diagram of a method **200** for operating a harvester is illustrated in accordance with aspects of the present subject matter. In general, the method **200** will be described herein with reference to the agricultural harvester **10** and related components described with reference to FIGS. **1-6**. However, it will be appreciated that the disclosed method **200** may be implemented with harvesters having any other suitable configurations and/or within systems having any other suitable system configuration. In addition, although FIG. **7** depicts steps performed in a particular order for purposes of illustration and discussion, the methods discussed herein are not limited to any particular order or arrangement. One skilled in the art, using the disclosures provided herein, will appreciate that various steps of the method disclosed herein can be omitted, rearranged, combined, and/or adapted in various ways without deviating from the scope of the present disclosure.

[0060] In various examples, the harvester can include an elevator assembly including an elevator extending between a proximal end portion and a distal end portion and a discharge assembly.

[0061] As shown in FIG. **7** at **(202)**, the method **200** may include conveying a material along the elevator in a direction from the proximate end portion to the distal end portion. As provided herein, when the harvester is configured as a sugarcane harvester, severed sugarcane stalks are conveyed via a feed roller assembly to a chopper assembly that cuts or chops the sugarcane stalks into pieces

or billets (e.g., six-inch cane sections). The processed harvested material discharged from the chopper assembly is then directed as a stream of billets and debris into a primary extractor, within which the airborne debris (e.g., dust, dirt, leaves, etc.) is separated from the sugarcane billets. The separated/cleaned billets then fall into the elevator assembly.

[0062] At **(204)**, the method can include receiving an input to alter the position of the discharge housing relative to the elevator housing from an input device. In various examples, the input device includes at least one of an input device, an internal data source, an external data source, or a sensor system.

[0063] At **(206)**, the method **200** can include altering a position of a discharge housing of the discharge assembly relative to an elevator housing with a discharge actuator. In some instances, a position of the deflector is based at least in part on a distance between the elevator and the elevator housing.

[0064] The discharge assembly downstream of the elevator may be configured to direct the harvested material in a defined direction and/or be used to assist in retaining the material within the elevator assembly for a defined time. In various examples, the discharge assembly can include a discharge housing that may be movable, pivotable, and/or translatable relative to the elevator housing. The discharge housing may define a discharge outlet. A deflector may be operably coupled with the discharge housing. A discharge actuator may be operably coupled with the elevator housing and the discharge housing.

[0065] At **(208)**, the method **200** can include rotating a deflector operably coupled with the discharge housing. In some instances, the deflector may be rotated based on the position of the discharge housing. Additionally or alternatively, at **(210)**, the method **200** can include rotating the deflector to a position in which at least a portion of the discharge outlet is blocked by the deflector to define a storage harvesting mode with a deflector actuator. Additionally, when in the storage harvesting mode, at **(212)**, the method **200** can include reducing a speed of a conveyor downstream of the elevator in the storage harvesting mode and/or, at **(212)**, the method **200** can include reducing an operating speed of the elevator in the storage harvesting mode. In an unloading mode, the deflector may be moved to a position in which at least a portion of the deflector is positioned vertically below the discharge housing.

[0066] In various examples, the method **200** may implement machine learning methods and algorithms that utilize one or several vehicle learning techniques including, for example, decision tree learning, including, for example, random forest or conditional inference trees methods, neural networks, support vector machines, clustering, and Bayesian networks. These algorithms can include computer-executable code that can be retrieved by the computing system and/or through a network/cloud and may be used to evaluate and update any performed processes. In some instances, the machine learning engine may allow for changes to be performed without human intervention.

[0067] It is to be understood that the steps of any method disclosed herein may be performed by a computing system upon loading and executing software code or instructions that are tangibly stored on a tangible computer-readable medium, such as on a magnetic medium, e.g., a computer hard drive, an optical medium, e.g., an optical disc, solid-state memory, e.g., flash memory, or other storage media known in the art. Thus, any of the functionality performed by the computing system described herein, such as any of the disclosed methods, may be implemented in software code or instructions that are tangibly stored on a tangible computer-readable medium. The computing system loads the software code or instructions via a direct interface with the computer-readable medium or via a wired and/or wireless network. Upon loading and executing such software code or instructions by the controller, the computing system may perform any of the functionality of the computing system described herein, including any steps of the disclosed methods.

[0068] The term “software code” or “code” used herein refers to any instructions or set of instructions that influence the operation of a computer or controller. They may exist in a computer-

executable form, such as vehicle code, which is the set of instructions and data directly executed by a computer's central processing unit or by a controller, a human-understandable form, such as source code, which may be compiled in order to be executed by a computer's central processing unit or by a controller, or an intermediate form, such as object code, which is produced by a compiler. As used herein, the term "software code" or "code" also includes any human-understandable computer instructions or set of instructions, e.g., a script, that may be executed on the fly with the aid of an interpreter executed by a computer's central processing unit or by a controller.

[0069] This written description uses examples to disclose the technology, including the best mode, and also to enable any person skilled in the art to practice the technology, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the technology is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. An elevator assembly for a harvester, comprising: an elevator housing; an elevator extending within the elevator housing between a proximal end portion and a distal end portion, the elevator configured to carry harvested crops between the proximal end portion and the distal end portion; and a discharge assembly operably coupled with the elevator housing, the discharge assembly comprising: a discharge housing defining a discharge outlet, the discharge housing movable relative to the elevator housing; and a deflector operably coupled with the discharge housing.
2. The elevator assembly of claim 1, further comprising: a discharge actuator operably coupled with the elevator housing and the discharge housing, the discharge actuator configured to alter a position of the discharge housing relative to the elevator housing.
3. The elevator assembly of claim 2, further comprising: a first link operably coupled with the elevator housing and the discharge actuator.
4. The elevator assembly of claim 2, further comprising: a second link operably coupled with the discharge housing and the discharge actuator.
5. The elevator assembly of claim 2, wherein the elevator housing defines an elevator opening, and wherein the discharge housing is positioned at least partially within the elevator opening.
6. The elevator assembly of claim 1, further comprising: a member operably coupled with the discharge housing; and a guide operably coupled with the elevator housing, wherein the guide is configured to direct a movement of the member along the guide.
7. The elevator assembly of claim 1, wherein the deflector may be positioned in a first position to direct harvested material in a defined direction downstream of the discharge housing and a second position to define a hopper.
8. The elevator assembly of claim 1, wherein the discharge housing is positioned at a first angle relative to the elevator housing in a first position and at a second angle relative to the elevator housing in a second position, the first angle varied from the second angle.
9. The elevator assembly of claim 2, further comprising: a computing system operably coupled with the discharge actuator; and an input device operably coupled with the computing system, wherein the computing system is configured to alter a position of the discharge housing relative to the elevator housing based on an input received through the input device.
10. A method for operating a harvester, the harvester comprising an elevator assembly including an elevator extending between a proximal end portion and a distal end portion and a discharge assembly, the method comprising: conveying a material along the elevator in a direction from the proximate end portion to the distal end portion; altering, with a discharge actuator, a position of a

discharge housing of the discharge assembly relative to an elevator housing, the discharge assembly downstream of the elevator; and rotating a deflector operably coupled with the discharge housing.

11. The method of claim 10, further comprising: receiving, from an input device, an input to alter the position of the discharge housing relative to the elevator housing.

12. The method of claim 11, wherein the input device includes at least one of a user interface, an internal data source, an external data source, or a sensor system.

13. The method of claim 10, further comprising: rotating, with a deflector actuator, the deflector to a position in which at least a portion of a discharge outlet defined by the discharge housing is blocked by the deflector to define a storage harvesting mode.

14. The method of claim 13, further comprising: reducing a speed of a conveyor downstream of the elevator in the storage harvesting mode.

15. The method of claim 13, further comprising: reducing an operating speed of the elevator in the storage harvesting mode.

16. The method of claim 10, wherein a position of the deflector is based at least in part on a distance between the elevator and the elevator housing.

17. An elevator assembly for a harvester, comprising: an elevator housing; an elevator configured to carry material there along; and a discharge assembly operably coupled with the elevator housing, the discharge assembly comprising: a discharge housing defining a discharge outlet, the discharge housing movable relative to the elevator housing; and a discharge actuator operably coupled with the elevator housing and the discharge housing, the discharge actuator configured to alter a position of the discharge housing relative to the elevator housing.

18. The elevator assembly of claim 17, further comprising: a deflector operably coupled with the discharge housing; and a deflector actuator configured to alter a position of the deflector relative to the discharge housing between a first position in which at least a portion of the discharge outlet is blocked by the deflector to define a storage harvesting mode and a second position in which at least a portion of the deflector is positioned vertically below the discharge housing.

19. The elevator assembly of claim 17, further comprising: a computing system operably coupled with the discharge actuator; and an input device operably coupled with the computing system, wherein the computing system is configured to alter a position of the discharge housing relative to the elevator housing based on an input received through the input device.

20. The elevator assembly of claim 19, wherein the input device includes at least one of a user interface, an internal data source, an external data source, or a sensor system.
